# Position-DETR: Step-by-Step Position-Guided Small Object Detection in Remote Sensing Images

Wenkai Zhao, Xinyu Deng, Fei Gao, and Chun Liu

***Abstract*—Detecting small objects in remote sensing images is highly challenging, primarily due to background noise interference and the limited semantic information of small objects. Although detection transformer (DETR)-like methods have significantly improved detection performance by introducing the transformer architecture, their performance remains suboptimal for small target detection. This is primarily because these methods struggle to effectively focus on small objects, leading to an equal treatment of extensive background information and sparse foreground information. This phenomenon not only results in the oversight of small objects but also introduces unnecessary computational burdens and slower convergence speeds. To address the above problems, we propose Position-DETR. First, we design a hierarchical foreground selection (HFS) module and a fine-grained foreground enhancement (FGFE) module to resolve computational bias and redundancy in the encoder. Furthermore, we propose progressive attention optimization (PAO) and hybrid matching query (HMQ) in the decoder to supplement small object information and improve convergence speed. Experimental results on four public remote sensing datasets for small target detection (DOTA-v1.0, DIOR, VISDRONE-2019, and STAR) demonstrate the effectiveness of Position-DETR, achieving average precision (AP) scores of 42.3%, 65.5%, 25.3%, and 21.3%, respectively, surpassing previous state-of-the-art results on each dataset. Our method achieves significantly faster convergence while maintaining low computational complexity. The source code will be available at https://github.com/wenkaizhao/Position-DETR**

***Index Terms*—Cross-attention, detection transformer (DETR), remote sensing images, small object detection, transformer.**

## I. Introduction

DRONE and satellite remote sensing technologies have been advancing rapidly and are widely utilized in various fields, including agriculture, urban planning, and environmental protection [1], [2], [3], [4], [5]. Consequently, the analysis of remote sensing images has gained increasing importance. However, the rich information contained in remote sensing images is often concentrated in very small areas (pixels less than $32 \times 32$), posing significant challenges for accurate small object detection. Although CNN-based methods, such as the RCNN series [6], [7], [8], YOLO series [9], [10], [11], and SSD series [12], [13], [14], have made significant progress, their local receptive fields restrict their ability to capture long-range dependencies. When handling extremely small or densely distributed small objects, the existing methods face ongoing challenges, including inadequate feature representation and severe background noise interference. Furthermore, CNN-based object detectors often rely on handcrafted components, such as nonmaximum suppression (NMS). These components not only introduce additional computational complexity but also significantly impair detection performance. Thus, exploring more efficient feature extraction mechanisms and model optimization strategies is crucial for improving the accuracy of small object detection.

In recent years, DETR [15], as a revolutionary end-to-end model, leverages the transformer architecture to enhance global information capture, addressing the limitations of CNN-based methods that rely on manually designed components. However, the global attention mechanism in DETR incurs high computational costs and struggles with small object detection when relying on high-level feature maps. Furthermore, compared to conventional detectors that allow multiple predictions to be assigned to each ground-truth box, the original DETR framework exclusively adopts a one-to-one matching strategy, where each ground truth is assigned to only one query. This results in most queries being excluded from regression training, thereby lacking sufficient supervisory signals for bounding box localization and category discrimination. Such sparse supervision compromises the quality of generated predictions and contributes to slow convergence during training. These limitations have motivated researchers to pursue further optimizations and enhancements. For instance, deformable-DETR [16] incorporates deformable attention based on deformable convolutions [17], focusing exclusively on a sparse set of key sampling points around reference positions. Although this approach achieves superior performance over the original DETR architecture, its emphasis on individual objects and their local contexts leads to insufficient global feature integration. Meanwhile, DINO [18] introduces contrastive denoising (CDN) training, which enhances discrimination capabilities by injecting heavily noised queries as negative samples to eliminate unproductive anchors. Although this innovation demonstrates performance surpassing CNN-like methods on

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COCO [19] benchmarks, similar to most DETR variants, its uniform computational allocation across all encoder tokens induces inherent computational bias and redundancy. Based on the above analysis, we can summarize three main problems of the current DETR-like methods.

1) In the encoder, the existing methods treat extensive background information and sparse foreground signals equally, resulting in both increased computational overhead and degradation of foreground feature distinctiveness.
2) For small objects, due to their limited semantic information, the existing methods have failed to effectively supplement contextual information, making it difficult for the model's attention to focus on small objects.
3) During the decoding stage, the majority of queries fail to participate in regression training and lack direct supervision from matched positive samples, which degrades the quality of the generated proposals and leads to slow convergence.

To address these challenges, we propose Position-DETR, a small object detection method for remote sensing images. The approach employs a hierarchical foreground selection (HFS) module to precisely screen detection object queries, complemented by a fine-grained foreground enhancement (FGFE) module to accurately amplify feature representations of small objects. These components work synergistically to progressively refine foreground focus while suppressing background interference, achieving enhanced detection accuracy with reduced computational overhead. In the decoder, we design progressive attention optimization (PAO) and hybrid matching query (HMQ) to improve the ability to discover potential small objects, thereby mitigating redundant predictions and accelerating model convergence. Our contributions are summarized as follows.

1) We propose a novel method for small object detection in aerial imagery. On the DOTA-v1.0 [20], DIOR [21], VISDRONE-2019 [22], and STAR [23] datasets, our method achieves average precision (AP) scores of 42.3, 65.5, 25.3, and 21.3, respectively, outperforming current state-of-the-art models.
2) We designed the HFS module to achieve fine screening of foreground tokens, effectively addressing computational redundancy and background interference issues.
3) We propose an FGFE module in the encoder and PAO in the decoder. These improvements not only strengthen semantic representation of small objects but also reduce redundant predictions.
4) We present an HMQ strategy that significantly strengthens query-positive sample matching capabilities while accelerating model convergence.

The remainder of this article is structured as follows. Section II presents the related work on small object detection and DETR-like methods. Section III details the overall architecture of the Position-DETR model and the proposed methodology. Section IV presents the experimental results. Section V concludes this article and summarizes the key contributions.

## II. Related Work

### A. Small Object Detection

Small object detection remains a profoundly challenging research topic in computer vision. The core difficulty stems from the minimal spatial coverage of small objects in images (typically spanning several to dozens of pixels), resulting in scarce visual features and ambiguous semantic representations. During feature extraction, conventional detection networks predominantly employ deep convolutional architectures. However, as network depth increases, critical details of small objects tend to dissipate through pooling or downsampling operations, while background noise interference further compromises their discriminative feature characteristics.

To address these challenges, current research primarily focuses on feature enhancement strategies. For mitigating the scarcity of visual features in small objects, Cui et al. [24] and Zhang et al. [25] have empirically validated the efficacy of super-resolution (SR) techniques in object detection. Li et al. [26] pioneered the integration of generative adversarial networks (GANs) [27] with feature SR methods, achieving significant performance improvements in small object detection. Bai et al. [28] devised a cooperative framework between generators and discriminators, where the generator specializes in augmenting image details, while the discriminator concurrently performs image authenticity verification, object classification, and precise localization. Noh et al. [29] proposed a super-resolution feature generator that enhances detection performance through high-resolution target feature supervision and relative receptive field matching between input–output features. Nevertheless, these methods have yet to achieve optimal equilibrium between feature authenticity and semantic consistency with detection objectives, while incurring substantial computational overhead. To address the challenge of constrained inference speed in edge computing environments, Zhang et al. [30] proposed APDNet, a lightweight framework that integrates knowledge distillation, low-precision quantization, and enhanced detection modules to achieve real-time high-accuracy aerial person detection on edge devices. Concurrently, Mei et al. [31] developed a stepwise activation quantization method for accelerating convolutional neural networks (CNNs), which constrains network inputs to low-bit integers to reduce memory footprint and accelerate inference.

Multiscale feature fusion constitutes another critical research direction. The introduction of the feature pyramid network (FPN) [32] enabled detection networks to simultaneously capture fine-grained details and global semantic information of small objects. Subsequent advancements, including NAS-FPN [33], ASFF [34], and BiFPN [35], have markedly improved detection performance. Building upon FPN, Liu et al. [36] proposed bottom–up path augmentation, which strengthens the propagation of shallow-layer details by introducing shortcut connections from low-level to high-level features. Guo et al. [37] introduced consistent supervision and residual feature augmentation mechanisms to enhance semantic coherence and detail preservation in multiscale features. To address noise amplification during FPN-based feature fusion [38], [39], [40],

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Liu et al. [41] developed a denoising FPN that suppresses noise during training while maintaining inference efficiency. Complementary approaches leverage contextual information surrounding small objects or their relational dependencies with more detectable counterparts to facilitate recognition. Mei et al. [42] enhanced image feature extraction capabilities by integrating spatial neighborhood statistical features (mean and standard deviation) to model contextual dependencies between objects. Bell et al. [43] developed the inside–outside net incorporating recurrent neural networks (RNNs) [44] to capture global context through four-directional scanning. Tang et al. [45] introduced Pyramidbox leveraging head-body co-occurrence priors to improve face detection accuracy through associated region reasoning. Hu et al. [46] proposed RelationNetwork using transformer-based relationship modeling to enhance feature representations for small objects. However, the aforementioned methods have failed to effectively address the issue of background noise interference, resulting in the persistent compromise of feature representation in detection targets.

### B. DETR-Like Methods

Detection transformer (DETR) was first introduced by Carion et al. [15] as a groundbreaking approach that integrates transformer architecture into object detection tasks. DETR simplifies the training process, eliminates the need for manually designed components, and removes the requirement for complex postprocessing steps. Despite its innovative design, DETR exhibits slow convergence, particularly during the early training stages, often requiring hundreds of epochs to match the performance of classical methods. Additionally, DETR's suboptimal performance in small object detection tasks has motivated subsequent research to optimize and enhance its capabilities.

Deformable-DETR incorporates multiscale feature maps and deformable attention modules, enabling the model to effectively capture target information across different scales and enhancing detection performance. DAB-DETR [47] significantly enhances the model's adaptability to varying object sizes and shapes by employing a flexible dynamic anchor box strategy that optimizes the number and placement of anchor boxes based on object characteristics. Sparse-DETR [48] employs a sparse sampling mechanism to select key locations for processing, significantly reducing computational load and improving efficiency. DN-DETR [49] enhances the model's robustness and generalization ability by introducing noise during training and further optimizes performance through noise reconstruction learning. DINO introduced a CDN training method and a hybrid query selection strategy for anchor box initialization in the decoder, establishing a benchmark for subsequent DETR-like model enhancements. DDQ [50] addresses the low recall rate and optimization challenges in dense object detection through a dense sparse query method, significantly improving detection performance. CO-DETR [51] enhances feature learning by incorporating a collaborative hybrid allocation training scheme and a customized positive sample query generation strategy. Align-DETR [52] mitigates the issue of output distribution mismatch during training by introducing an IoU-aware BCE loss function. MS-DETR [53] employs a mixture of one-to-one and one-to-many supervision to explicitly guide the candidate object generation process, thereby improving training efficiency. Relation-DETR [54] addresses the issue of inconspicuous interobject relationships in detection tasks by explicitly modeling spatial dependencies between objects. Frozen-DETR [55] significantly enhances detection performance by leveraging a frozen pretrained base model as a feature enhancer within the detector.

Existing approaches primarily focus on local object regions while inadequately addressing feature representation in encoders, often overemphasizing query optimization in decoders. This paradigm tends to compromise feature learning for small objects. In contrast, our proposed approach not only effectively suppresses background interference but also holistically leverages global object information, thereby providing more comprehensive and precise informational enhancement for small objects.

## III. Methodology

### A. Overview

The overall architecture of Position-DETR is illustrated in Fig. 1, with DINO serving as the baseline. Position-DETR is an end-to-end detector comprising a CNN backbone, an encoder, a decoder, and multiple prediction heads. The core design of Position-DETR leverages key positional information, including the foreground ground truth and the bounding boxes predicted by each decoder layer, to effectively capture local and global features, enhance small object detection performance, and improve model convergence. Specifically, the input image is first processed by the backbone network to extract multiscale features, which are then fed into the HFS. The filtered tokens are subsequently passed to the transformer encoder. In the encoder, we incorporate the FGFE to refine feature enhancement. Subsequently, the top-K features are selected to initialize the object queries for the transformer decoder. These queries are fed into the decoder and integrated with the HMQ training strategy. In the decoder's self-attention mechanism, one-to-one matching is optimized through PAO, guided by position-aware priors at each layer, while one-to-many matching facilitates query learning to match positive samples. Sections III-B–III-E provide detailed descriptions of the four modules.

### B. Hierarchical Foreground Selection

Spares-DETR and Focus-DETR [56] have demonstrated that utilizing partial foreground information as queries in the encoder can achieve comparable detection performance. However, for remote sensing images with complex backgrounds, accurately selecting foreground information that occupies minimal regions remains a challenge. As illustrated in Fig. 2, Focus-DETR exhibits deficiencies in foreground selection, particularly in images with significant variations in target sizes, where it tends to overlook small-sized objects. We attribute this primarily to the insufficiently strict supervision in the selection process. We propose a scale-agnostic and

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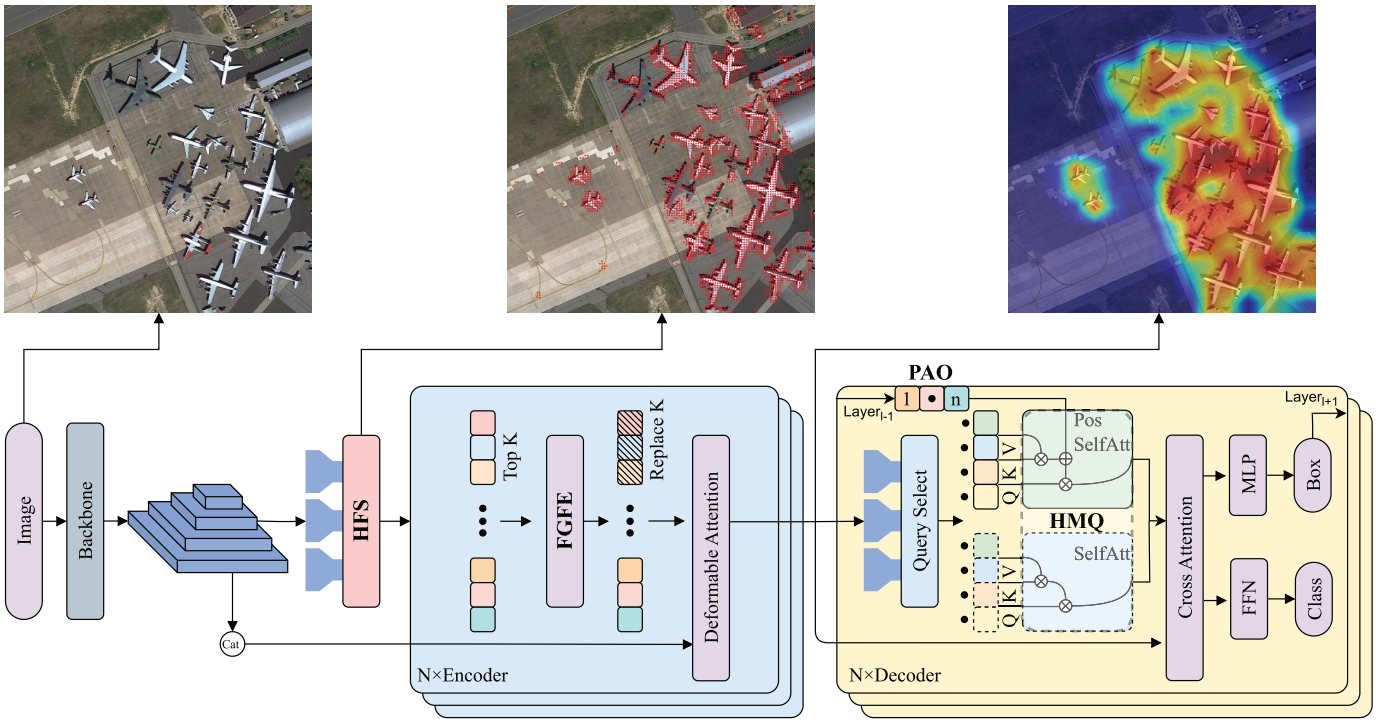



Fig. 1. Overview of the proposed Position-DETR model. The model comprises three key components: a backbone network, a transformer encoder, and a transformer decoder. To enhance the feature extraction capability of the encoder, we introduce HFS module after the backbone network and FGFE module within the encoder. Furthermore, we incorporate PAO module into the decoder and incorporates HMQ module on top of the one-to-one matching mechanism to address the issues of matching instability and slow convergence.

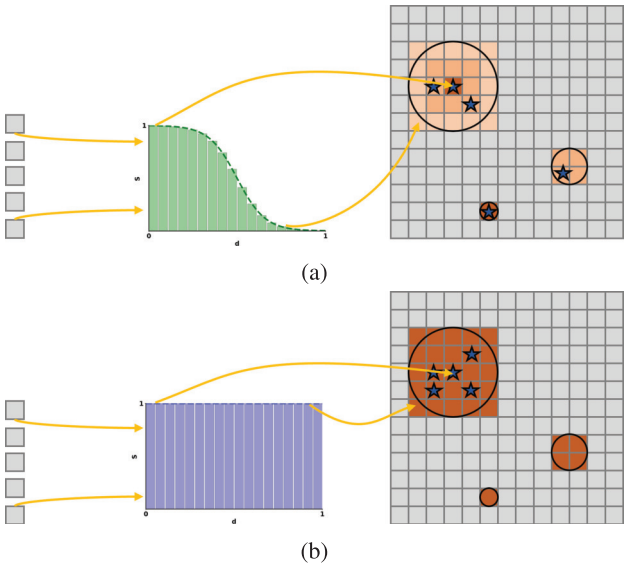



Fig. 2. Comparison between (a) our proposed method and (b) focus-DETR in foreground score supervision. The gray squares represent the feature map tokens extracted through the backbone network, where the color intensity corresponds to the confidence scores of tokens being foreground objects. Asterisk markers indicate the top five candidate queries selected as having the highest probability of being foreground.

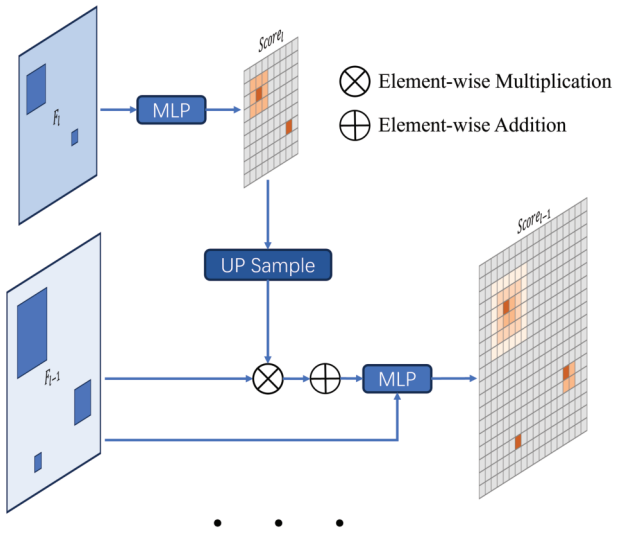



Fig. 3. Structure of the foreground scoring module.

more precise selection and supervision mechanism to mitigate scale bias.

Specifically, using the backbone network and downsampling operations, we extract four multiscale feature maps $\{f_1, f_2, f_3, f_4\}$, where $f_4$ is derived from downsampling $f_3$. These feature maps are fed sequentially from high to low resolution ($f_4 \rightarrow f_1$) into a foreground scoring module, which consists of a multilayer perceptron (MLP), to predict the confidence score of each token as foreground. Fig. 3 illustrated the structure of the foreground scoring module. Our scoring design explicitly considers the complementary characteristics of different feature levels: high-level feature maps exhibit rich semantic information but suffer from low resolution, making them less effective for small target detection. In contrast, low-level feature maps retain more spatial details but lack sufficient

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semantic information. Similar to the top–down approach in the FPN, we leverage the foreground scores from high-level semantic feature maps to guide the foreground token scores of adjacent low-level feature maps. For a feature map $f_l$, where $l \in \{2, 3, 4\}$

$$S_{l-1} = \mathbf{MLP}_{\mathrm{F}}\left(f_{l-1}\left(1 + \mathbf{UP}\left(\alpha_l * S_l\right)\right)\right) \tag{1}$$

where $S_l$ denotes the foreground score of the $l$th layer feature map, UP denotes the bilinear upsampling function, $\mathrm{MLP_F}$ is a global score predictor, and $a_{l=1}^{l-1}$ is a set of learnable modulation coefficients. This establishes an interconnection between the foreground score predictions of different feature maps.

To guide this foreground query scoring across multiscale features, we develop a foreground position supervision module to guide query scoring across multiscale features. Specifically, the ground truth is projected onto each feature map. For each token $T_l^{(i,j)}$ at position $(i, j)$ in the $l$th feature map, it corresponds to the coordinates $c = (x, y)$ in the original image. Our approach introduces finer-grained supervision: the guidance score for a token as foreground is dynamically adjusted based on its distance from the ground-truth center, represented as the confidence score of the foreground query $\mathrm{Score}_l^{(i,j)}$, which is calculated using the following formula:

$$\mathrm{Score}_l^{(ij)} = \begin{cases} S\left(P_x, P_y\right), & \left(P_x, P_y\right) \in \mathrm{DBbox} \\ 0, & \left(P_x, P_y\right) \notin \mathrm{DBbox} \end{cases} \tag{2}$$

where $\mathrm{DBbox} = (x, y, w, h)$ denotes the true bounding box, and $(p_x, p_y)$ represents the position coordinates of each token $T_l^{(i,j)}$. The score of $T_l^{(i,j)}$ as a foreground is determined by its relative distance to the target center

$$S\left(P_x, P_y\right) = \frac{1 - \tanh\left[2\pi\left(\frac{d\left(P_x, P_y\right)}{d_{\max}} - \frac{1}{2}\right)\right]}{2} \tag{3}$$

where $d(P_x, P_y)$ denotes the Euclidean distance between the coordinates $(P_x, P_y)$ and the center point $(C_x, C_y)$, and $d_{\max}$ denotes the maximum distance from the center point to the edge of the ground-truth bounding box.

To enhance multiscale feature processing, an encoder-level filtering is designed in addition to the feature layer-by-layer filtering. We specify the feature-level filtering ratio $\{v_t\}_{t=1}^{\mathrm{T}}$ and the encoder-layer filtering ratio $\{w_l\}_{l=1}^{L}$, where $T$ and $L$ denote the total number of feature layers and encoder layers, respectively. For the encoder-layer filtering ratio, in the initial stages, more queries are retained to capture global information. As the depth of the encoder layers increases, the filtering ratio is progressively decreased to eliminate redundant information, enabling the model to focus on local features and better capture fine-grained contextual details. Following encoder-layer query filtering, the computational cost of the deformable attention mechanism is significantly reduced from:

$$O\left(\sum_{l=1}^{L} H_l W_l C T\left(C + KC + 5K + 3MK\right)\right) \tag{4}$$

to

$$O\left(\sum_{1 \le l \le L} \sum_{1 \le t \le T} w_l v_t H_l W_l C\left(C + KC + 5K + 3MK\right)\right) \tag{5}$$

where $M$ and $K$ represent the number of attention heads and the number of sampled keys, respectively.

### C. Fine-Grained Foreground Enhancement

The tokens filtered by the HFS module contain precise and rich information about the detection targets. Effectively leveraging these critical tokens is essential for improving the accuracy of small object detection. Inspired by decoder architectures, we introduce a self-attention mechanism prior to the deformable attention module within the encoder to achieve fine-grained feature refinement. This module facilitates interaction between different object queries, captures inter-object relationships, and enables the prelocalization of target positions before the cross-attention stage, thereby enhancing the representational capacity of foreground features. The core design principle of the FGFE module is to leverage the candidate tokens, dynamically selected by the HFS module for their salient semantic features. Through cross-object relationship modeling and enhanced context awareness, refined representation of small object features within the encoder.

**Algorithm 1** FGFE

**Input:** Images $X \in \mathbb{R}^{H \times W \times C}$, Position embedding $PE$, Object token number $k_s$, Encoder layer count $L$, Encoder layer filtering ratio $\{w_l\}_{l=1}^{L}$

**Output:** Memory

$F_1, F_2, F_3 = \mathrm{Backbone}(X)$, $F_4 = \mathrm{downsample}(F_3)$
**for** each scale $i \in \{1, 2, 3, 4\}$ **do**
  $T_i \leftarrow \mathrm{Flatten}(F_i)$
**end for**
$T_a \leftarrow \mathrm{Concat}([T_1, T_2, T_3, T_4])$
$S_f \leftarrow \mathrm{HFS}(T_a)$
**for** layer $n = 1$ to $L$ **do**
  $N_{\mathrm{total}} \leftarrow \mathrm{len}(T_a)$
  $k_l \leftarrow \lceil w_l \cdot N_{\mathrm{total}} \rceil$
  $\mathrm{idx}_l \leftarrow \mathrm{TopK}(S_f, k_l)$
  $T_l \leftarrow T_a[\mathrm{idx}_l]$, $S_l \leftarrow S_f[\mathrm{idx}_l]$
  $\mathrm{Idx}_s \leftarrow \mathrm{TopK}(S_l, k_s)$
  $T_s \leftarrow T_l[\mathrm{idx}_s]$
  $T_s \leftarrow \mathrm{MHSA}(T_s + PE, T_s + PE, T_s)$
  $T_l[\mathrm{idx}_s] \leftarrow T_s$
  $\mathrm{Memory} \leftarrow \mathrm{MSDeformAttn}(T_l + PE, T_a + PE, T_a)$
**end for**
**return Memory**

Specifically, the input image undergoes feature extraction via a backbone network, with all feature maps concatenated to form the initial token set $T_a$. The HFS module calculates foreground scores $S_f$ for each token in $T_a$. At the $l$th encoder layer, candidate queries $T_l$ and their corresponding scores $S_l$ are first selected based on global scores $S_f$ and the layer-specific filtering ratio $w_l$. Subsequently, the top-$k_s$ high-scoring tokens from $T_l$ are selected as target tokens $T_s$. These undergo multihead self-attention (MHSA) with positional embeddings to model interobject relationships and reinforce semantic coherence. Finally, the refined tokens are reintegrated into $T_l$ and processed through multiscale deformable

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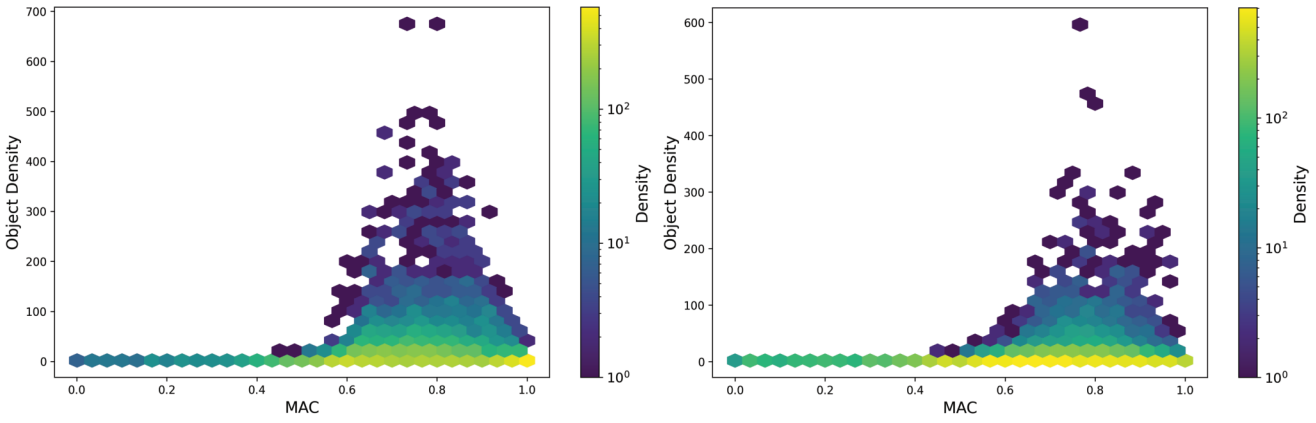



Fig. 4. Visualization results of object relationships in (a) DOTA-v1.0 dataset and (b) DIOR dataset.

attention (MSDeformAttn) to generate enhanced features. The detailed procedure of the FGFE algorithm is formalized in Algorithm 1.

### D. Progressive Attention Optimization

Early research has demonstrated that modeling interobject relationships can significantly enhance the performance of detectors. To mitigate the issue of slow convergence, most DETR-like methods, such as Deformable-DETR and DN-DETR, predominantly focus on individual objects and their local contexts while neglecting global information, thereby limiting the model's ability to comprehend the entire scene. Due to the limited spatial coverage and information provided by small objects, effectively leveraging global target information in the image becomes crucial to supplement their characteristics. To evaluate the correlation between objects within images, we visualized the object relationships in the DOTA-v1.0 dataset [Fig. 4(a)] and the DIOR dataset [Fig. 4(b)] based on Pearson correlation coefficients. We computed the mean absolute correlation (MAC) using ground-truth information $b = (x, y, w, h)$ from images to quantify the linear correlation degree among objects, formulated as follows:

$$r_{ij} = \frac{\operatorname{cov}\left(\mathbf{b}_i, \mathbf{b}_j\right)}{\sigma\left(\mathbf{b}_i\right)\sigma\left(\mathbf{b}_j\right)} \tag{6}$$

$$\mathrm{MAC} = \frac{2}{n(n-1)} \sum_{1 \le i < j \le n} \left|r_{ij}\right| \tag{7}$$

where $r$ denotes the Pearson correlation coefficient and $n$ represents the number of objects.

When MAC = 1, it indicates complete linear correlation among all objects; when MAC = 0, it signifies the absence of linear correlation between any pair of objects. In the figure, colors closer to yellow represent a higher count of images within that interval. The figure shows that the MAC values of images in the dataset are predominantly clustered in the higher range, indicating that there is a statistically significant linear relationship between the objects.

Inspired by Relation-DETR, we construct high-dimensional relation embeddings from the spatial and geometric information of predicted bounding boxes. These embeddings serve as prior information to guide the self-attention mechanism within the transformer decoder. In contrast, our proposed modeling approach is better suited for small object detection in remote sensing. Specifically, we guide the attention mechanism of the current layer by modeling the relationships based on the prediction results of the preceding layer, and we explicitly integrate relative spatial relationships into the relation representation, thereby enhancing both the relevance and richness of the modeled information. To ensure scale invariance and improve numerical stability, we employ normalized relative spatial and geometric features to characterize the various relationships. The mathematical expression of our relation modeling method is as follows:

$$R = \begin{cases} e\left(b_i, b_j\right), & i \neq j \\ 0, & i = j \end{cases} \tag{8}$$

where $e(b_i, b_j)$ represents the relationship modeling between different predicted bounding boxes, where $i, j \in N$ denotes the top-$k$ queries in the decoder. $b = (x, y, w, h)$ refers to the predicted bounding box output by the decoder

$$e\left(b_i, b_j\right) = \left[\alpha, \beta, \gamma, \delta\right] \tag{9}$$

$$\alpha = \operatorname{sign}\left(x_i - x_j\right) \cdot \log\left(\frac{\left|x_i - x_j\right|}{w_i} + 1\right) \tag{10}$$

$$\beta = \operatorname{sign}\left(y_i - y_j\right) \cdot \log\left(\frac{\left|y_i - y_j\right|}{h_i} + 1\right) \tag{11}$$

$$\gamma = \log\left(\frac{w_i}{w_j}\right) \tag{12}$$

$$\delta = \log\left(\frac{h_i}{h_j}\right). \tag{13}$$

The relationship matrix $\mathbf{E} \in \mathbb{R}^{N \times N \times 4}$ is then further transformed into a high-dimensional embedding using sinusoidal positional encoding. The sign function captures the relative directional relationships between objects

$$\operatorname{Embed}\left(\mathbf{R}, 2k\right) = \sin\left(\frac{s\mathbf{R}}{T^{2k/d_{\mathrm{re}}}}\right) \tag{14}$$

$$\operatorname{Embed}\left(\mathbf{R}, 2k+1\right) = \cos\left(\frac{s\mathbf{R}}{T^{2k/d_{\mathrm{re}}}}\right). \tag{15}$$

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The shape of the relational embedding is $N\times N\times 4d_{\mathrm{re}}$, where $T$, $s$, and $d_{\mathrm{re}}$ are the encoding parameters. Finally, the embedding undergoes a linear transformation to obtain $M$ scalar weights, where $M$ represents the number of attention heads

$$\mathrm{Pos}\left(\boldsymbol{b}_i,\boldsymbol{b}_j\right)=\max\left(\epsilon,\mathbf{W}\mathrm{Embed}\left(\boldsymbol{b}_i,\boldsymbol{b}_j\right)+\mathbf{B}\right) \tag{16}$$

where $\epsilon$ is introduced to ensure that Pos remains positive, preventing gradient vanishing. $\mathrm{Pos}(b,b)\in R^{N\times N\times M}$. We then use the relationship weights of the current layer to optimize the attention mechanism of the next encoder layer

$$\mathrm{AttnScore}\left(\mathbf{Q}^l\right)=\mathrm{Pos}^{l-1}+\frac{\mathrm{Que}\left(\mathbf{Q}^l\right)\mathrm{Key}\left(\mathbf{Q}^l\right)^{\mathrm{T}}}{\sqrt{d_{\mathrm{model}}}} \tag{17}$$

$$\mathrm{Attn}_{\mathrm{Self}}\left(\mathbf{Q}^l\right)=\mathrm{Softmax}\left(\mathrm{AttnScore}\left(\mathbf{Q}^l\right)\right)\mathrm{Val}\left(\mathbf{Q}^l\right) \tag{18}$$

where $Q^l$ represents the query of the $l$th layer in the encoder, and $\mathrm{Pos}^{l-1}$ denotes the relationship weights from the $(l-1)$th layer of the encoder.

### E. Hybrid Matching Query

The traditional DETR architecture employs a one-to-one matching mechanism: each ground-truth bounding box is matched to a single query, and all remaining queries are considered background. This mechanism causes the majority of queries to fail to participate in regression training and to lack direct supervision for matching positive samples. Consequently, the predicted bounding boxes become jittery and unstable, making it difficult to ensure their quality. Furthermore, the PAO module relies on predicted bounding boxes for relation modeling to optimize the decoder's self-attention mechanism. Therefore, the quality of the predicted bounding boxes not only directly impacts model performance but also indirectly affects the accuracy of global relation modeling. Existing methods, such as MS-DETR and MR-DETR [57], mitigate this issue by selecting the top six highest scoring queries for each target when computing the loss function. However, in remote sensing object detection tasks involving small objects, dense scenes are common (often exceeding 300 targets per image). In such scenarios, assigning a fixed number of queries to each target may result in some targets not being effectively queried, leading to reduced recall. To address this issue, we propose the HMQ module. During training, while retaining the one-to-one matching mechanism, we introduce an auxiliary one-to-many matching mechanism. This enables more queries to associate effectively with ground-truth objects, providing richer and denser supervision signals. This enhances the positive sample matching capability and ensures the quality of candidate boxes. During inference, only the one-to-one branch is retained, preserving the end-to-end nature of the original DETR and not introducing additional inference costs.

Specifically, we present two types of parallel queries: one-to-one matching queries $Q=\{q_1,q_2,\ldots,q_m\}$ and one-to-many mixed queries $\widetilde{Q}=\{q_1,q_2,\ldots,q_h\}$. During training, both query sets are fed in parallel into the decoder's self-attention mechanism, and an attention mask is employed to prevent information leakage between the two query sets. For the one-to-one matching queries $Q$, we incorporate the positional relationships of predicted bounding boxes to enhance the model's ability to eliminate duplicate predictions. For the one-to-many mixed queries $\widetilde{Q}$, positional relationship information is omitted to facilitate the exploration of unknown objects. For one-to-one matching, the predictions $P=\{p_1,p_2,\ldots,p_m\}$ from each decoder layer are aligned with the ground truth $G=\{g_1,g_2,\ldots,g_N\}$ after $L$ decoder layers, and the loss $\mathcal{L}_m$ is computed

$$\mathcal{L}_m(P,G)=\sum_{l=1}^{L}\mathcal{L}_{\mathrm{Hungarian}}(P,G). \tag{19}$$

Similarly, the same number of $L$ decoder layers is used to process the mixed queries, resulting in predictions $\widetilde{P}=\{p_1,p_2,\ldots,p_h\}$. The ground truth $G$ is replicated k times to obtain $\widetilde{G}=\{g_i^k\mid i\in[N],k\in[K]\}$, enabling one-to-many matching. After each layer's prediction, one-to-many matching is performed, and the loss $\mathcal{L}_h$ is calculated

$$\mathcal{L}_h\left(\tilde{P},\tilde{G}\right)=\sum_{l=1}^{L}\mathcal{L}_{\mathrm{Hungarian}}\left(\tilde{P},\tilde{G}\right). \tag{20}$$

During training, the total loss for mixed matching is $\mathcal{L}_m+\mathcal{L}_h$. In the inference phase, only the one-to-one matching of the original DETR model is retained, ensuring no additional computational overhead during inference.

### F. Optimization

Similar to other DETR-like methods, our loss function is defined as follows:

$$\mathcal{L}_{\mathrm{total}}=\lambda_m\mathcal{L}_m+\lambda_{\mathrm{dn}}\mathcal{L}_{\mathrm{dn}}+\lambda_{\mathrm{enc}}\mathcal{L}_{\mathrm{enc}}+\lambda_f\mathcal{L}_f+\lambda_h\mathcal{L}_h \tag{21}$$

where $\mathcal{L}_m$ is the matching loss based on the Hungarian algorithm, $\mathcal{L}_{\mathrm{dn}}$ is the denoising loss, $\mathcal{L}_f$ is the foreground selection loss, $\mathcal{L}_{\mathrm{enc}}$ is the auxiliary optimization loss for the encoder output, and $\mathcal{L}_h$ is the loss for mixed matching

$$\mathcal{L}_f=-\alpha_f\left(1-p_f\right)^{\gamma}\log\left(p_f\right) \tag{22}$$

where $\mathcal{L}_f$ represents the supervised loss function in the HFS module, and $\alpha_f=0.25$ and $\gamma=2$ are the hyperparameters for foreground selection, respectively.

## IV. Experiments

### A. Dataset

To evaluate the effectiveness of our method, we conducted extensive experiments on three widely used datasets. The characteristics of each dataset are described as follows.

1) *DOTA-v1.0:* The DOTA-v1.0 is a recognized benchmark dataset for remote sensing object detection tasks. It comprises 2806 high-resolution remote sensing images, ranging in size from 800 × 800 to 4000 × 4000 pixels, covering diverse scenarios, such as urban, rural, and industrial areas. The dataset encompasses 15 categories, including airplanes, vehicles, and buildings, which are of significant practical importance in remote sensing applications. To mitigate information loss caused by object segmentation during preprocessing, we employed the official image slicing tool to crop the original images into 1024 × 1024 subimages

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with an overlap of 200 pixels. This process yielded 15 749 training images and 5292 validation images.

2) *DIOR:* The DIOR is a challenging remote sensing object detection dataset aimed at detecting multiscale objects in complex backgrounds. It comprises 23 463 remote sensing images with a total of 190 288 annotated object instances. All images have a resolution of 800 × 800 pixels and encompass ten primary categories, including buildings, vehicles, and swimming pools. These categories represent typical remote sensing object scenarios, exhibiting high diversity and generality. The dataset is partitioned into 19 004 training images and 2347 validation images.
3) *VISDRONE-2019:* The VISDRONE-2019 dataset emphasizes object detection tasks from a drone perspective, featuring high dynamic range and complex scene backgrounds. It includes 288 video sequences with 261 908 video frames and 10 209 static images. For our experiments, we utilized the 10 209 static images, with resolutions ranging from 960 × 540 to 1920 × 1080 pixels. The dataset encompasses 12 categories of common objects, including pedestrians, vehicles, bicycles, and tricycles. The static images are further partitioned into 6741 training images and 548 validation images.
4) *STAR:* The STAR dataset is a large-scale scene graph generation (SGG) benchmark designed for high-resolution satellite imagery. It contains over 210 000 annotated objects and 400 000 relational triplets, spanning 11 complex geographic scene categories including airports, ports, and other critical infrastructure. With image resolutions ranging from 512 × 768 to 27 860 × 31 096 pixels, the dataset supports multiscale object detection and spatial relationship analysis across 48 fine-grained object categories, such as ships, vehicles, aircraft, and bridges. We used an image-slicing tool to crop the original images into 1024 × 1024 pixel patches with a 200-pixel overlap. This process produced 70 383 training samples and 32 641 validation samples.

### B. Implementation Details

The implementation details of Position-DETR and other comparative methods are aligned with those of other DETR-like approaches. We employ the AdamW optimizer for training on an NVIDIA RTX 3060 GPU (12GB). The weight decay is configured to $1 \times 10^{-4}$, with the backbone network's initial learning rate set to $1 \times 10^{-5}$ and other components' initial learning rate set to $1 \times 10^{-4}$. The learning rate is reduced by a factor of 0.1 at the 11th epoch. The batch size per GPU is configured to 1. The parameters $T$, $d_{\text{re}}$, and $s$ are determined empirically as 10 000, 16, and 100, respectively. The filtering ratios for the feature layers and encoder layers were set to (0.4, 0.8, 1.0, 1.0) and (1.0, 0.8, 0.6, 0.6, 0.4, 0.2), respectively. The mixed training configuration follows the settings established by DINO and other DETR variants employing one-to-many matching, with $N_m = 900$, $N_h = 1500$, and $k = 6$. The image preprocessing utilizes the same augmentation pipeline as in other DETR-based detectors, comprising random resizing, cropping, and flipping. The comparative experiments are conducted using the Detrex [58] and MMDetection [59] frameworks.

### C. Evaluation Criteria

We evaluated model performance using eight metrics based on the MS COCO standard: AP, $AP_{50}$, $AP_{75}$, $AP_S$, $AP_M$, $AP_L$, parameters (Params), and giga floating-point operations (GFLOPs). Specifically, AP is defined as the mean precision across ten intersection-over-union (IoU) thresholds ranging from 0.5 to 0.95 with 0.05 increments, mathematically expressed as follows:

$$\text{AP} = \int_0^1 P(R)\, dR \tag{23}$$

where $P$ and $R$ denote the precision and recall, respectively, calculated as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{24}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{25}$$

where TP, FP, and FN represent true positives, false positives, and false negatives, respectively. Generally, AP is conventionally used to assess overall detection performance and robustness. $AP_S$, $AP_M$, and $AP_L$ quantify detection accuracy for small ($< 32^2$ pixels), medium ($32^2$–$96^2$ pixels), and large ($\geq 96^2$ pixels) targets, respectively. Params indicate model size, and GFLOPs measure computational complexity.

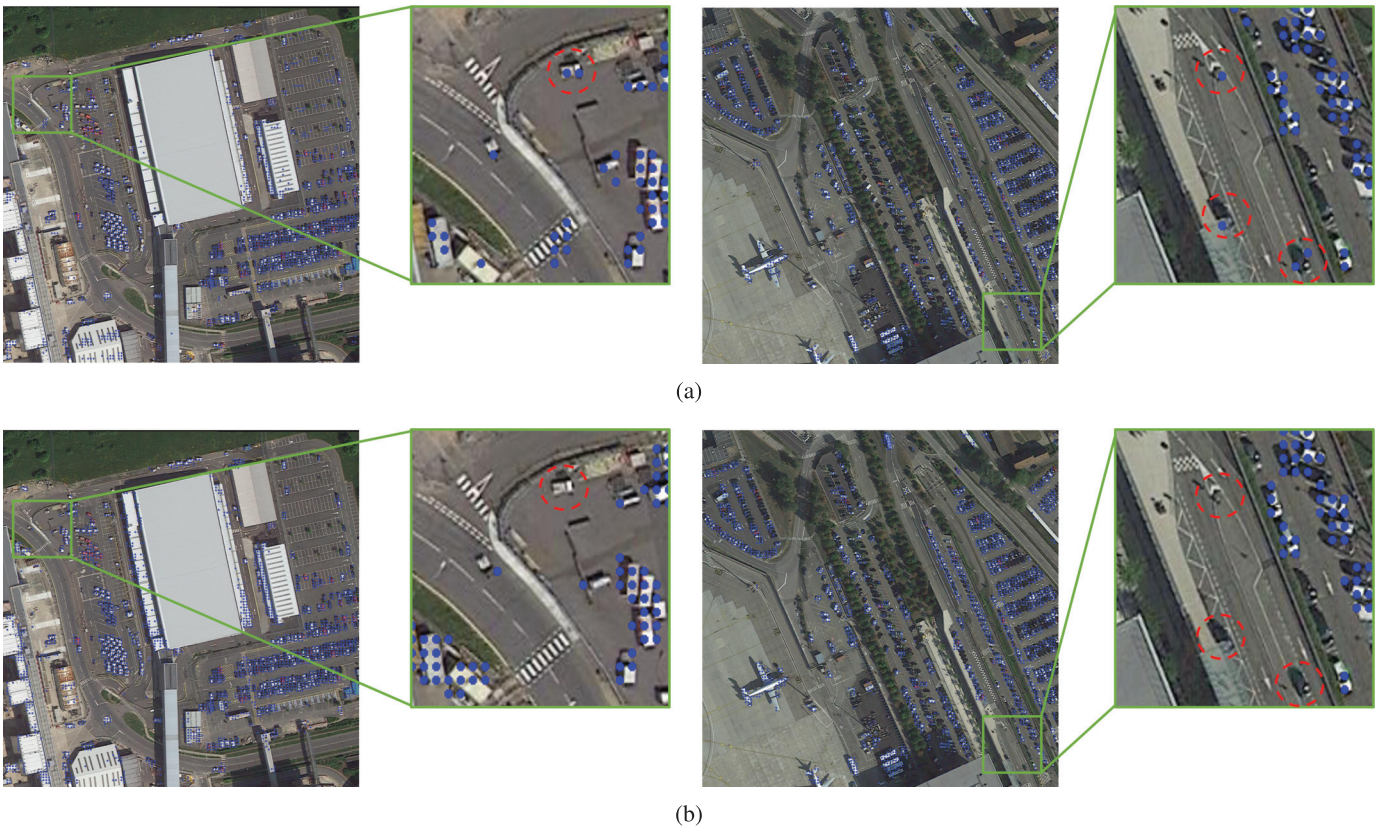
### D. Comparison With the State-of-the-Art Methods

We experimentally evaluate Position-DETR on four datasets: DOTA-v1.0, DIOR, VISDRONE-2019, and STAR. Fig. 5 presentsthe visualization of detection results from the Position-DETR model on the DOTA-v1.0 dataset. Experimental results demonstrate that Position-DETR achieves the highest detection accuracy across all datasets compared to the state-of-the-art detection models, including DDQ, H-DINO, Salience-DETR, and Focus-DETR. We conducted a comparative analysis between Position-DETR and other advanced DETR variants, with detailed results presented in Table I. Position-DETR attains the highest detection accuracy (AP of 42.3%) while requiring only 184 GFLOPs of computational cost. Compared to the baseline DINO model with a ResNet-50 [62] backbone, the introduction modules increase the parameter count by a modest 5.05%. However, this modest parameter overhead is outweighed by substantial gains in performance and efficiency. Importantly, Position-DETR yields a 16.85% improvement in detection accuracy while concurrently reducing computational complexity by over 35%. Compared to Focus-DETR, our approach incurs a 12.2% increase in computational complexity but delivers superior performance, boosting both AP and $AP_S$ by 8.74% and 16.94%, respectively. Notably, accuracy gains are especially pronounced for small object detection. Position-DETR effectively addresses computational bias and redundancy during training, thereby enhancing detection accuracy and achieving a strong balance between precision and computational efficiency.

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Fig. 5. Examples of detection results on the DOTA-v1.0 dataset using Position-DETR.



(a)

(b)

Fig. 6. Visualization results of foreground query selection in (a) our method and (b) focus-DETR.

Distinct from methods, such as DDQ and Align-DETR, which directly utilize multiscale features as encoder queries, this mechanism often results in feature confusion when processing small targets in complex scenes and incurs high computational costs. In contrast, Position-DETR outperforms in handling small and overlapping targets due to our proposed HFS and PAO. Key features are preserved and enhanced during the encoder stage, while the query information of target regions is further refined in the decoder stage, significantly improving small object

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TABLE I

RESULTS FOR OUR POSITION-DETR AND OTHER DETECTION MODELS WITH THE RESNET-50 BACKBONE ON DOTA-V1.0. THE FIRST AND SECOND BEST RESULTS ARE MARKED IN RED AND BLUE, RESPECTIVELY

| Model | Epochs | Backbone | AP | $AP_{50}$ | $AP_{75}$ | $AP_S$ | $AP_M$ | $AP_L$ | Params | GFLOPs |
|---|---|---|---|---|---|---|---|---|---|---|
| Faster-RCNN [6] | 12 | ResNet50 | 29.7 | 55.1 | 28.1 | 11.2 | 32.1 | 37.0 | 41.7M | 209 |
| Cascade-RCNN [7] | 12 | ResNet50 | 32.0 | 56.2 | 32.0 | 11.9 | 33.8 | 40.5 | 69.3M | 237 |
| YOLOv5 | 100 | CSPDarknet | 37.7 | 58.5 | 40.9 | 17.7 | 38.7 | 48.4 | 2.5M | 7.2 |
| YOLOv11 | 100 | CSPDarknet | **39.5** | 61.3 | 42.9 | 17.6 | 39.5 | **50.8** | 2.6M | 6.5 |
| Deformable-DETR [16] | 12 | ResNet50 | 32.7 | 57.1 | 33.0 | 12.4 | 33.9 | 44.6 | 40.4M | 207 |
| DAB-Def-DETR [47] | 12 | ResNet50 | 27.6 | 51.5 | 26.4 | 11.4 | 32.2 | 34.4 | 43.6M | 246 |
| H-Def-DETR [60] | 12 | ResNet50 | 37.3 | 62.8 | 37.9 | 14.3 | 39.0 | 48.9 | – | – |
| Salience-DETR [61] | 12 | ResNet50 | 33.8 | 53.6 | 35.0 | 18.2 | 34.2 | 42.6 | 55.9M | 202 |
| H-DINO [60] | 12 | ResNet50 | 33.0 | 56.8 | 35.7 | 16.5 | 36.7 | 42.3 | 47.6M | 299 |
| Focus-DETR [56] | 12 | ResNet50 | 38.9 | 63.7 | 40.8 | 18.3 | 40.8 | 49.7 | 49.3M | 164 |
| DINO [18] | 12 | ResNet50 | 36.2 | 62.2 | 39.4 | 18.8 | 39.4 | 46.0 | 47.5M | 284 |
| DDQ [50] | 12 | ResNet50 | 38.2 | 65.5 | 43.0 | 20.3 | 40.9 | 51.3 | – | – |
| Align-DETR [52] | 12 | ResNet50 | 39.3 | **66.2** | **43.7** | **20.6** | **41.1** | 48.6 | 47.5M | 280 |
| Position-DETR(ours) | 12 | ResNet50 | **42.3** | **66.6** | **45.3** | **21.4** | **43.6** | **53.4** | 49.9M | 184 |
| Salience-DETR [61] | 24 | ResNet50 | 35.1 | 55.2 | 37.6 | 17.6 | 35.5 | 42.8 | 55.9M | 202 |
| DINO [18] | 24 | ResNet50 | 40.3 | **66.2** | **44.9** | **21.4** | **43.7** | 51.0 | 47.5M | 284 |
| Focus-DETR [56] | 24 | ResNet50 | **41.0** | 65.8 | 43.6 | **18.9** | 42.8 | **52.8** | 49.3M | 164 |
| Position-DETR(ours) | 24 | ResNet50 | **43.1** | **67.2** | **46.4** | **21.4** | **44.2** | **53.6** | 49.9M | 184 |

detection performance without increasing computational complexity.

Compared to H-DINO, both methods utilize one-to-one and one-to-many matching mechanisms. However, in specific scenarios, such as small object detection, when small object features are not sufficiently attended to, naively increasing the number of queries may cause query matching conflicts. As demonstrated in Table I, H-DINO achieves lower detection accuracy compared to DINO. Our method effectively addresses this issue, enabling hybrid matching strategy to overcome the limitations of small object detection scenarios and fully leverage the advantages of one-to-many matching.

To validate the robustness and effectiveness of our method, we conducted further evaluations on the DIOR, VISDRONE-2019, and STAR datasets, with the corresponding results presented in Tables II–IV, respectively. In addition, we quantified the proportional distribution of objects of different scales within each dataset to analyze the impact of scale distribution on detection accuracy, as detailed in Table V. Experimental results demonstrate that our method achieves performance improvements across all object scales. Specifically, large-scale objects typically exhibit more accurate and stable geometric priors due to occupying more pixels and having clearer contours. In scenarios where the scale distribution is relatively balanced, our method can leverage the precise semantic context provided by large objects to assist in the recognition of small objects, thereby significantly enhancing small-object detection performance. Moreover, for scale categories with limited samples in the dataset (e.g., large-scale objects in VISDRONE-2019), our method

TABLE II

RESULTS FOR OUR POSITION-DETR AND OTHER DETECTION MODELS WITH THE RESNET-50 BACKBONE ON DIOR. THE FIRST AND SECOND BEST RESULTS ARE MARKED IN RED AND BLUE, RESPECTIVELY

| Model | AP | $AP_{50}$ | $AP_{75}$ | $AP_S$ | $AP_M$ | $AP_L$ |
|---|---|---|---|---|---|---|
| Faster-RCNN [6] | 43.4 | 71.6 | 45.5 | 12.6 | 41.0 | 54.7 |
| Cascade-RCNN [7] | 48.6 | 75.1 | 53.4 | 13.3 | 45.0 | 61.5 |
| Sparse-RCNN [8] | 53.6 | 76.3 | 58.3 | 14.2 | 44.7 | 69.4 |
| DAB-Def-DETR [47] | 54.1 | 76.5 | 58.8 | 16.4 | 47.3 | 70.3 |
| DINO [18] | 57.6 | 79.7 | 62.4 | 21.7 | 50.7 | 72.3 |
| H-Def-DETR [60] | 63.1 | 85.7 | 68.4 | 26.2 | 55.5 | 77.8 |
| DN-Def-DETR [49] | 56.1 | 77.7 | 60.6 | 16.1 | 48.8 | 72.1 |
| H-DINO [60] | 57.0 | 78.3 | 62.5 | 20.4 | 50.0 | 72.0 |
| Focus-DETR [56] | 63.4 | 85.4 | 69.2 | 25.8 | 54.9 | 77.8 |
| Salience-DETR [61] | 61.0 | 82.5 | 65.6 | 24.4 | 54.4 | 76.1 |
| DDQ [50] | **64.7** | **87.9** | **71.0** | **26.4** | **57.6** | **79.4** |
| Position-DETR(ours) | **65.5** | **87.2** | **71.4** | **26.7** | **57.7** | **80.4** |

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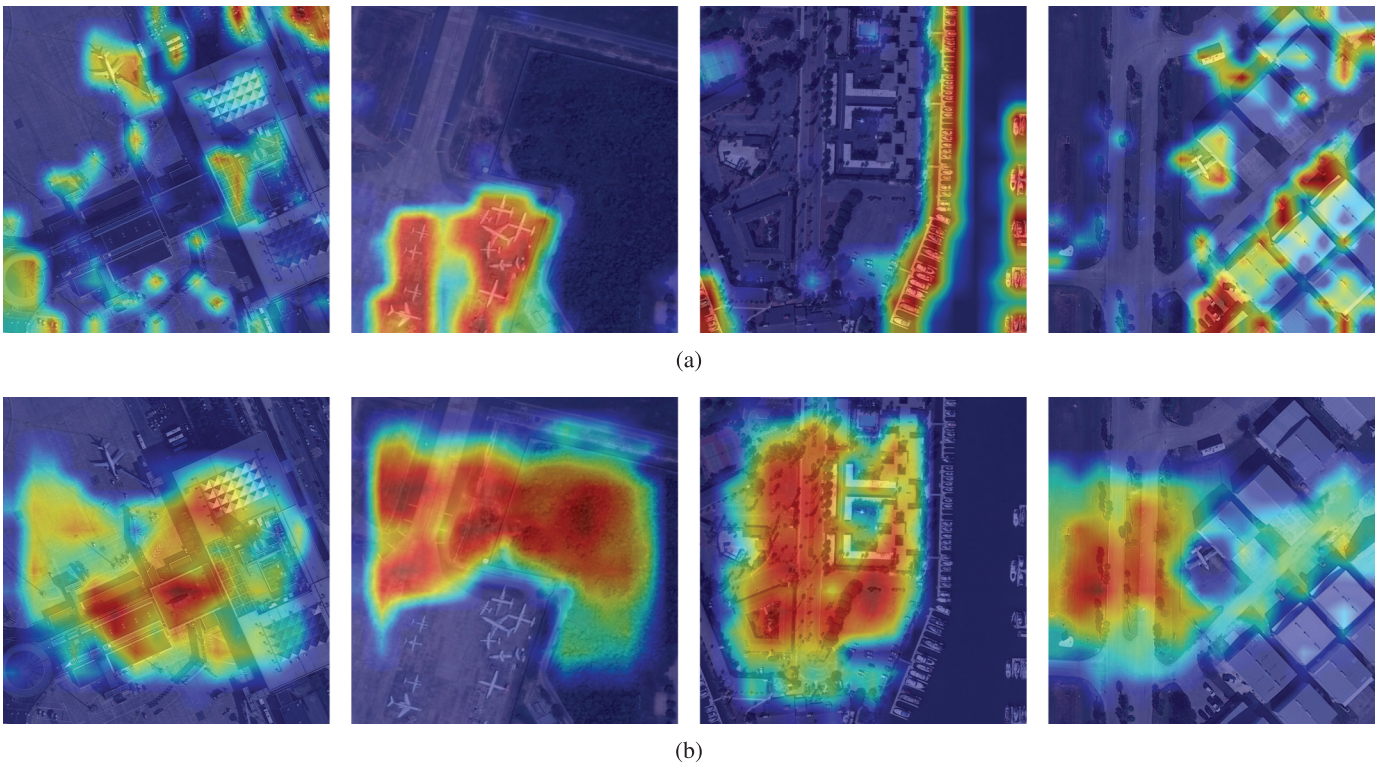


Fig. 7. GradCAM visualization results of encoder outputs in (a) our method and (b) DINO.

TABLE III

RESULTS FOR OUR POSITION-DETR AND OTHER DETECTION MODELS WITH THE RESNET-50 BACKBONE ON VISDRONE-2019. THE FIRST AND SECOND BEST RESULTS ARE MARKED IN RED AND BLUE, RESPECTIVELY

| Model | AP | $AP_{50}$ | $AP_{75}$ | $AP_S$ | $AP_M$ | $AP_L$ |
|---|---|---|---|---|---|---|
| Cascade-RCNN [7] | 18.3 | 29.9 | 19.3 | 11.5 | 26.9 | 28.3 |
| Faster-RCNN [6] | 16.6 | 28.7 | 17.3 | 10.5 | 25.2 | 27.3 |
| H-DINO [60] | 23.5 | 40.8 | 23.4 | 16.4 | 32.1 | 40.2 |
| Salience-DETR [61] | 23.7 | 40.9 | 23.4 | 15.6 | 32.8 | 45.6 |
| DDQ [50] | **24.6** | 41.6 | **24.9** | 16.5 | **33.3** | **41.5** |
| DINO [18] | 24.2 | **41.9** | 24.0 | **16.9** | 32.9 | 40.1 |
| Position-DETR(ours) | **25.3** | **43.3** | **25.4** | **17.1** | **34.4** | **48.1** |

TABLE IV

RESULTS FOR OUR POSITION-DETR AND OTHER DETECTION MODELS WITH THE RESNET-50 BACKBONE ON STAR. THE FIRST AND SECOND BEST RESULTS ARE MARKED IN RED AND BLUE, RESPECTIVELY

| Model | AP | $AP_{50}$ | $AP_{75}$ | $AP_S$ | $AP_M$ | $AP_L$ |
|---|---|---|---|---|---|---|
| Faster-RCNN [6] | 11.9 | 26.7 | 8.8 | 4.9 | 6.8 | 11.6 |
| Group-DETR [63] | 9.7 | 19.6 | 7.7 | 1.7 | 6.1 | 10.1 |
| H-Def-DETR [60] | 13.2 | 23.4 | 12.2 | 5.4 | 8.0 | 12.4 |
| Def-DETR [16] | 13.6 | 23.7 | 13.2 | 6.7 | 8.3 | 12.6 |
| Salience-DETR [61] | 16.0 | 28.5 | 15.2 | 8.7 | 10.4 | 15.6 |
| DINO [18] | **20.2** | **34.0** | **20.3** | **9.1** | **12.9** | **19.6** |
| Position-DETR(ours) | **21.3** | **36.2** | **21.4** | **11.4** | **14.0** | **21.0** |

can integrate abundant contextual information from other scales to enhance the feature representation of these scarce objects, thus alleviating the limitations of sample scarcity and substantially improving their detection accuracy.

Fig. 6 compares the foreground token selection results between our method and Focus-DETR. The selection mechanism of Focus-DETR lacks refinement and struggles to identify sparsely distributed small targets, such as vehicles scattered on roads. Our method resolves this limitation by preventing redundant selection of easily detectable foreground tokens and prioritizing independent objects. This improvement stems from our fine-grained scoring supervision mechanism, which ensures effective selection of small objects, including those at minimal scales.

To further validate the FGFE module, we apply GradCAM [64] to visualize attention heatmaps of encoder output features, comparing our method with DINO. As shown in Fig. 7, DINO's attention is predominantly concentrated on regions with dense object clusters under complex background interference, failing to distribute attention to isolated targets. Consequently, this leads to reduced detection accuracy. In contrast, our method accurately focuses on minuscule objects, thereby achieving superior detection performance.

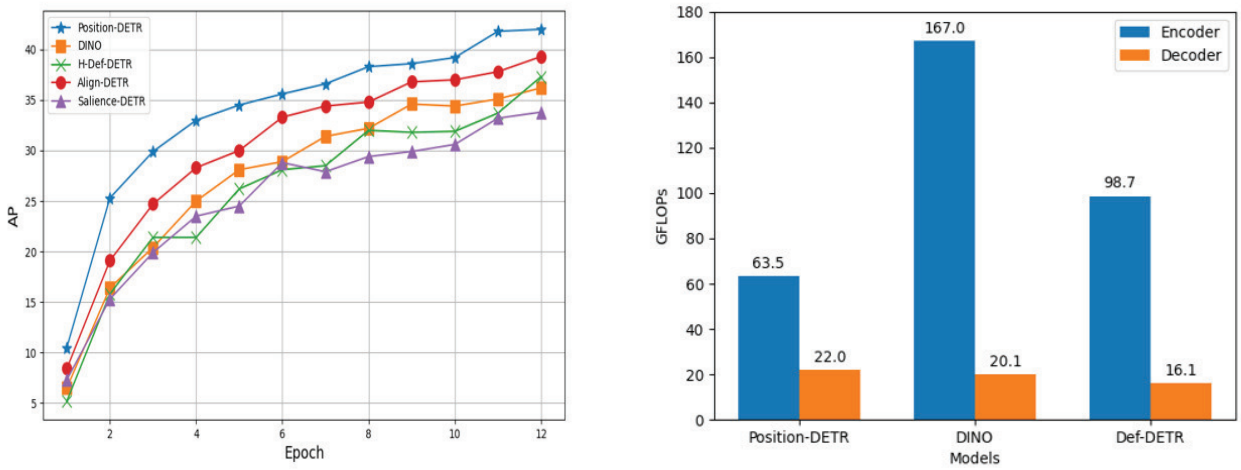
For a comprehensive evaluation of accuracy, convergence capability, and computational efficiency, Fig. 8(a) compares the AP values of four high-performance DETR variants and our method over the first 12 training epochs, whereas Fig. 8(b) quantifies the computational costs in the encoder

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TABLE V

COMPARISON OF $AP_S$/$AP_M$/$AP_L$ BETWEEN POSITION-DETR AND THE DINO BASELINE ON FOUR DATASETS. SIZE DIST INDICATES THE PROPORTION OF SMALL/MEDIUM/LARGE OBJECTS IN EACH DATASET

| Dataset | Size dist | $AP_S$ | | $AP_M$ | | $AP_L$ | |
|---|---|---|---|---|---|---|---|
| | | Baseline | Ours | Baseline | Ours | Baseline | Ours |
| DOTA-v1.0 | 39.4 / 48.4 / 12.2 | 18.8 | 21.4(+2.6) | 39.4 | 43.6(+4.2) | 46.0 | 53.4(+7.4) |
| DIOR | 51.6 / 29.9 / 18.5 | 21.7 | 26.7(+5.0) | 50.7 | 57.7(+7.0) | 72.3 | 80.4(+8.1) |
| VisDrone-2019 | 67.7 / 29.3 / 3.0 | 16.9 | 17.1(+0.2) | 32.9 | 34.4(+1.5) | 40.1 | 48.1(+8.0) |
| STAR | 40.2 / 33.3 / 26.5 | 9.1 | 11.4(+2.3) | 12.9 | 14.0(+1.1) | 19.6 | 21.0(+1.4) |




(a) (b)

Fig. 8. Comparative analysis of convergence curves and computational cost between our method and other DETR-like models, where all models use ResNet-50 as the backbone and are trained on the DOTA-v1.0 dataset with the same training configuration. (a) Training convergence curves comparison. (b) Computational cost comparison.

TABLE VI

EXPERIMENTAL RESULTS OF EACH PROPOSED COMPONENT ON DOTA-V1.0 DATASET

| HFS | PAO | HMQ | FGFE | AP | $AP_{50}$ | $AP_{75}$ | $AP_S$ | $AP_M$ | $AP_L$ |
|---|---|---|---|---|---|---|---|---|---|
| × | × | × | × | 36.2 | 62.2 | 39.4 | 18.8 | 39.4 | 46.0 |
| ✓ | × | × | × | 40.8 | 65.0 | 43.1 | 19.7 | 42.0 | 51.5 |
| ✓ | ✓ | × | × | 41.5 | 66.2 | 43.4 | 20.1 | 42.6 | 51.7 |
| ✓ | ✓ | × | ✓ | 41.9 | 66.5 | 44.0 | 20.8 | 43.2 | 52.0 |
| ✓ | ✓ | ✓ | × | 42.0 | 66.2 | 44.3 | 20.5 | 42.9 | 52.4 |
| ✓ | ✓ | ✓ | ✓ | **42.3** | **66.6** | **45.3** | **21.4** | **43.6** | **53.4** |

and decoder stages across three methods. Our approach consistently achieves the highest detection accuracy per epoch while significantly reducing encoder computational costs, attaining the state-of-the-art performance with minimal resource consumption.

### E. Ablation Study

To evaluate the effectiveness of each component, we perform ablation studies by integrating the four proposed modules into DINO. Our goal is to enhance small object detection in remote sensing images, with experiments performed on the DOTA-v1.0 dataset. Table VI summarizes the impact of each module on evaluation metrics, where ✓ indicates module inclusion and × denotes exclusion.

With the integration of the HFS module, the AP was improved by 4.6%, and the small-object AP ($AP_S$) increased by 0.9%. This improvement stems from the characteristic of remote sensing imagery, where background regions typically dominate the scene. The HFS module effectively suppresses

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background interference, allowing the encoder to precisely concentrate on small targets while enhancing their discriminative features. After incorporating PAO into the decoder, the AP further improved by 0.7%, demonstrating that PAO effectively utilizes prediction boxes during decoding and supplements small object information. To validate the auxiliary roles of the FGFE and HMQ modules, we added the FGFE module alone to a baseline consisting of the HFS and PAO modules. AP and $AP_S$ increased by 0.4% and 0.7%, respectively, verifying the effectiveness of the FGFE module in performing fine-grained feature interactions on the selected foreground tokens. Similarly, when individually incorporating the HMQ module into the retained HFS and PAO module baseline, AP and $AP_S$ improved by 0.5% and 0.4%, respectively, demonstrating that the introduction of the auxiliary one-to-many matching mechanism enhances the query's ability to match positive samples, thereby improving detection performance. With the integration of all four proposed modules, AP and $AP_S$ were improved by 6.1% and 2.6%, respectively, compared to the DINO baseline. The experimental results demonstrate that each component of the proposed algorithm contributed significantly to improve the model performance.

## V. Conclusion

In this article, we present Position-DETR, an innovative method for small object detection in remote sensing images, which addresses the limitations of existing DETR variants in this domain. We design four core modules (HFS, FGFE, PAO, and HMQ) to improve detection accuracy in complex backgrounds. The HFS module improves positional awareness of small targets while suppressing background interference and computational redundancy. The FGFE module strengthens discriminative feature extraction for small targets, enhancing their detectability in cluttered environments. The PAO module incorporates position-aware priors into the decoder, effectively suppressing redundant predictions and facilitating faster convergence. The HMQ module enhances the ability of queries to match positive samples, particularly in complex scenarios, such as dense or occluded environments, effectively mitigating false negatives caused by missed detections. Extensive experiments on three benchmark datasets (DOTA-v1.0, DIOR, VISDRONE-2019, and STAR) demonstrate that Position-DETR achieves the state-of-the-art performance, outperforming the existing methods by significant margins. Although this study has achieved significant progress in the domain of remote sensing small object detection, it still exhibits the following limitations.

1) While the proposed method substantially reduces computational costs compared to the existing DETR variants, its computational efficiency remains inferior to single-stage detectors.
2) The current experiments were conducted exclusively on three datasets, which remain insufficient to cover the rich application scenarios of remote sensing imagery, particularly in the more challenging task of spaceborne remote sensing image object detection.

With the advancement of remote sensing technologies, future research will focus on enhancing the feature representation of small objects through multimodal data integration, thereby enabling precise detection of extremely small targets in more complex scenarios.

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